

軟體工程在金融領域的轉移：情緒分析從傳統 ML 微調到 LLM 異質代理人與口頭強化機制的系統設計

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參考論文：

Designing Heterogeneous LLM Agents for Financial Sentiment Analysis(2024)

Sentiment trading with large language models(2024)

Outline

ML → LLM

- 動機與趨勢：金融 AI 的範式轉移
- Case Study (1): Sentiment trading with large language models
- 情緒分析傳統金融 ML 的限制：為何已不敷使用？
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動機與趨勢：金融 AI 的範式轉移

- 非結構化數據的複雜性激增：現代金融新聞充滿了隱喻、反諷和依賴上下文的語義。
- 對可解釋性的需求：論文中HAD 框架展示了如何透過代理人之間的「對話」，將推理過程顯性化。解決了對 AI 黑箱決策的擔憂。

X

傳統Loughran-McDonald Master Dictionary

"The company's loss narrowed significantly compared to last quarter."
(公司的虧損較上季大幅收窄。)

金融 NLP

代表模型： Loughran-McDonald Master Dictionary (LM Dictionary)。

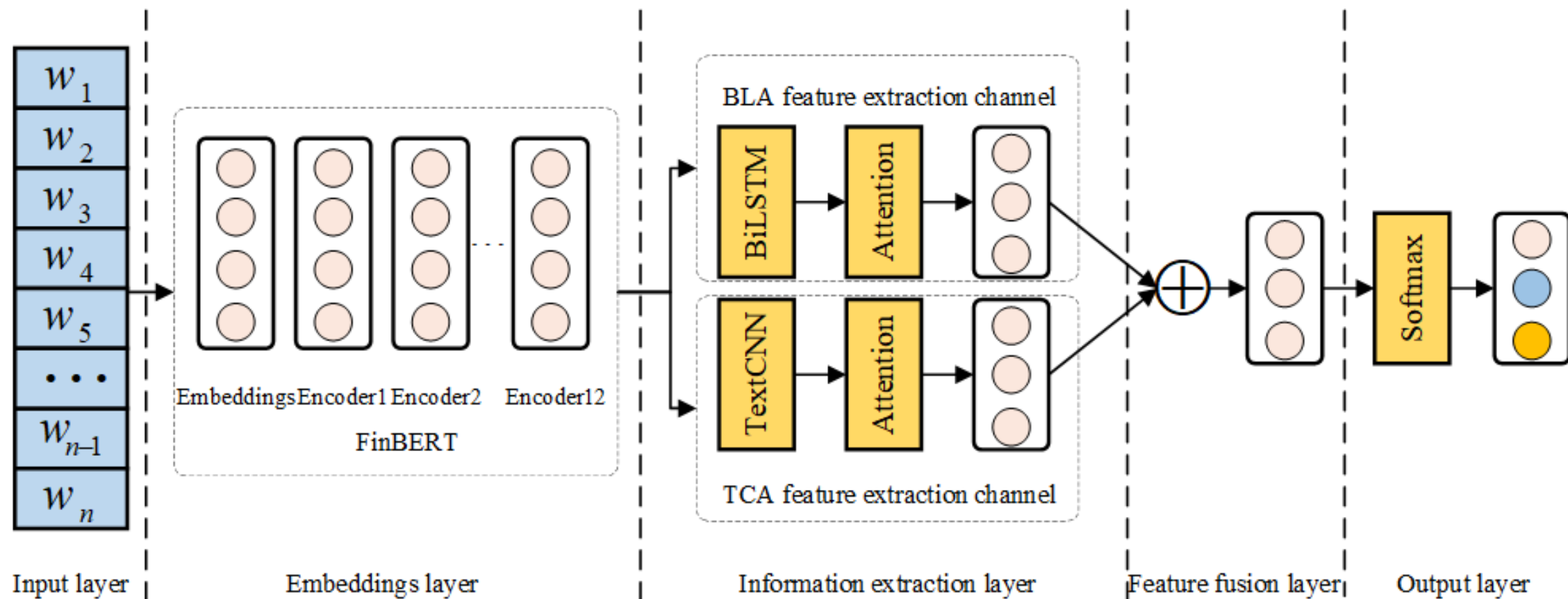
運作原理：

將文本視為一堆單詞的集合，計算特定金融情緒詞彙（如 "loss", "claim"）出現的頻率。

缺點：忽略了詞序和上下文關係。例如，它無法區分 "loss narrowed" (虧損收窄，好事) 和 "huge loss" (巨大虧損，壞事)。

誤以為是壞的

傳統Fin-BERT



Case Study (1): Sentiment trading with large language models

Table 3: Language model performance metrics: accuracy, precision, recall, specificity, and the F1 score for each model. The OPT model is the most accurate, followed closely by BERT and FINBERT.

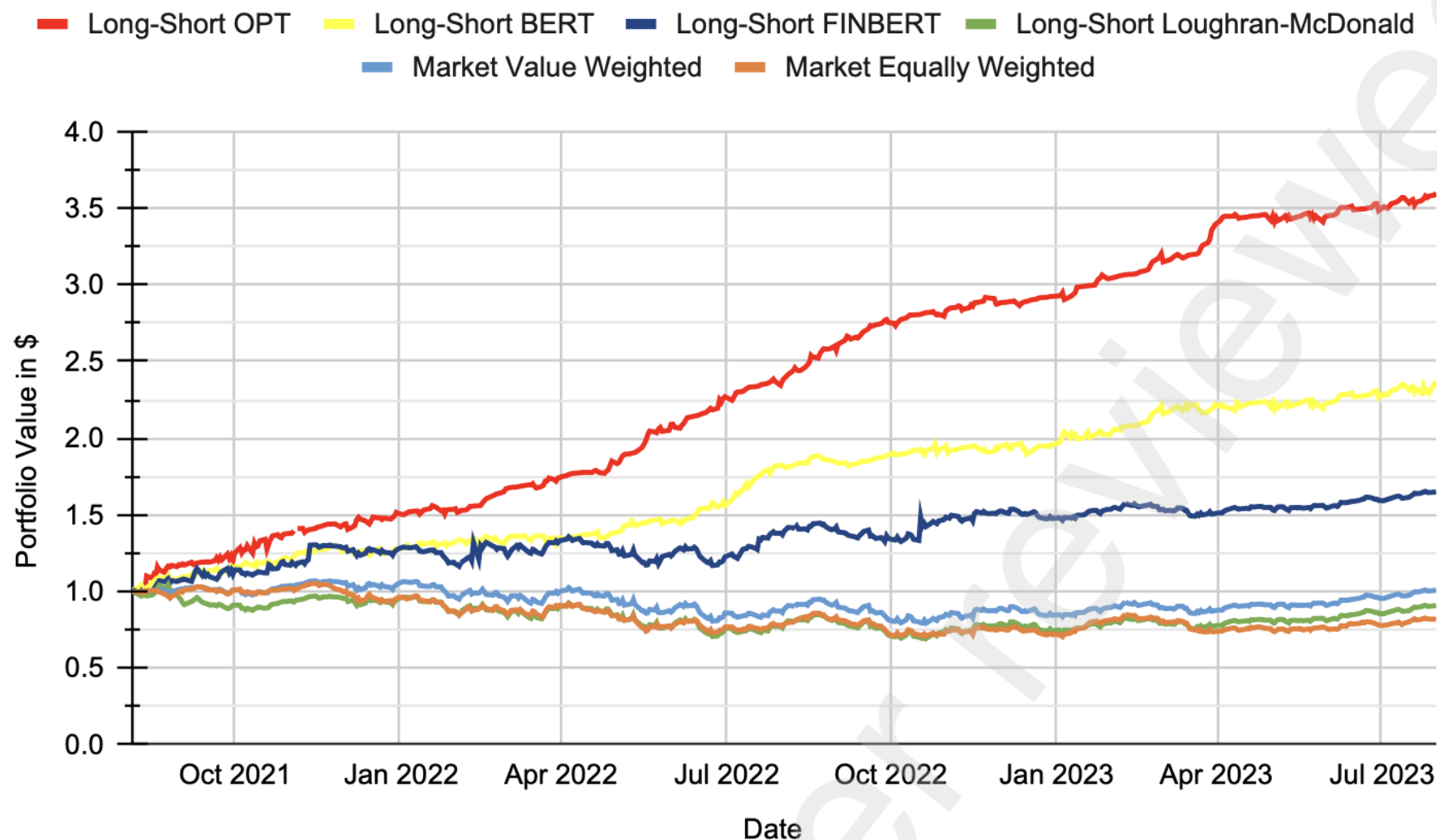


Metric	OPT	BERT	FINBERT	Loughran-McDonald
Accuracy	0.744	0.725	0.722	0.501
Precision	0.732	0.711	0.708	0.505
Recall	0.781	0.761	0.755	0.513
Specificity	0.711	0.693	0.685	0.522
F1 score	0.754	0.734	0.731	0.508

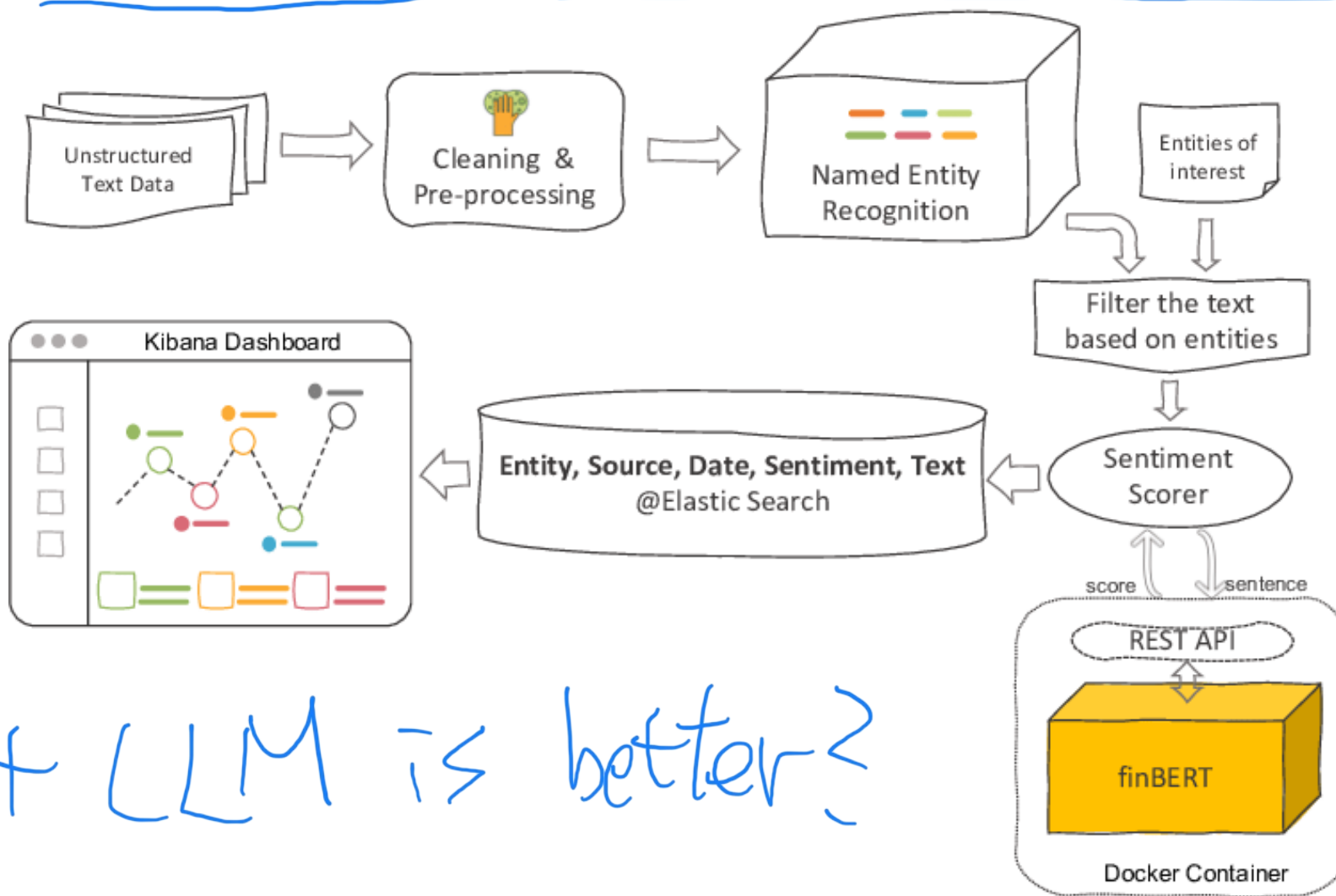
Case Study (1): Sentiment trading with large language models

	OPT			BERT			FinBERT		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Sharpe ratio	1.81	1.42	3.05	1.59	1.28	2.11	1.51	1.19	2.07
MDR (%)	0.32	0.25	0.55	0.25	0.21	0.45	0.22	0.18	0.39
StdDev (%)	2.18	2.91	2.49	2.49	3.19	2.68	2.59	3.31	2.81
MDD (%)	-14.76	-24.69	-18.57	-17.89	-27.95	-21.95	-19.71	-29.94	-23.82
	LM dictionary			EW			VW		
	Long	Short	L-S	Long	Short	L-S	Long	Short	L-S
Sharpe ratio	0.87	0.66	1.23	1.25	1.05	1.40	1.28	1.08	1.45
MDR (%)	0.12	0.13	0.22	0.18	0.15	0.33	0.19	0.16	0.35
StdDev (%)	3.54	4.13	3.74	2.90	3.70	3.20	2.95	3.75	3.25
MDD (%)	-35.47	-45.39	-38.29	-31.13	-42.21	-32.87	-28.76	-38.95	-31.87

Case Study (1): Sentiment trading with large language models

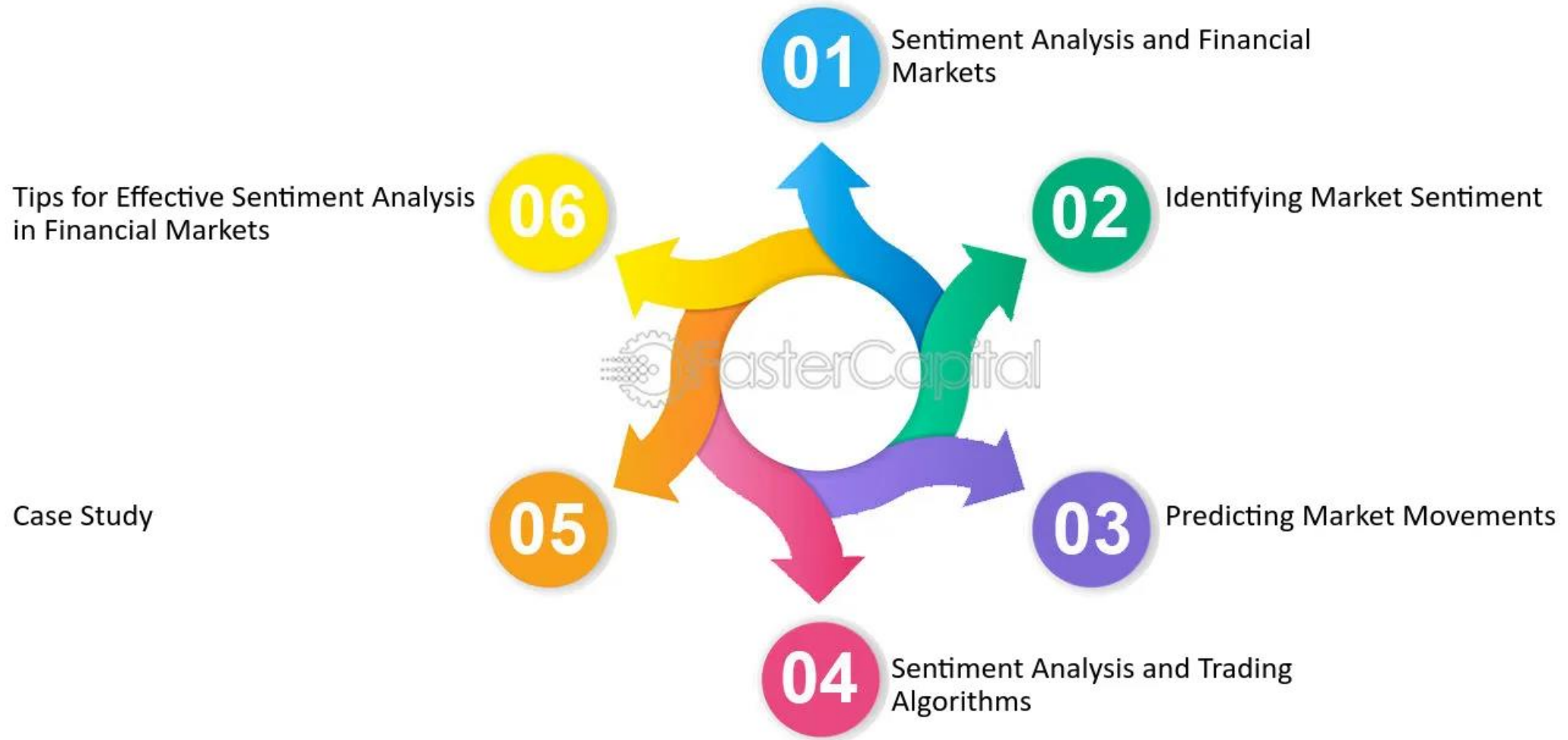


傳統金融 ML 的限制：為何已不敷使用？



ML + LLM is better?

Paradigm Shift: LLM 多代理系統崛起



Case Study (2): Designing Heterogeneous LLM Agents for Financial Sentiment Analysis

代理人 ID	名稱	Prompt 指令核心 (角色設定)	認知功能 (對應 Minsky 理論)
A1	語氣代理人 (Mood Agent)	"特別注意任何虛擬語氣 (irrealis mood) 的使用。"	區分事實與推測/謠言
A2	修辭代理人 (Rhetoric Agent)	"特別注意任何修辭（如反諷、雙重否定）。"	解碼語言複雜性與隱喻
A3	依賴性代理人 (Dependency Agent)	"專注於發言者的情緒，而非第三方引用。"	釐清情緒歸屬對象
A4	方面代理人 (Aspect Agent)	"專注於特定的股票代碼/主題，忽略其他實體。"	實體消歧義
A5	參照代理人 (Reference Agent)	"特別注意時間表達、價格及未言明的事實。"	數值與時間錨定
A6	機構投資者 (Institutional Agent)	"像機構投資者一樣思考，關注長期基本面。"	基本面分析模擬
A7	散戶投資者 (Individual Agent)	"像散戶一樣思考，關注價格波動與技術指標。"	市場情緒與動能模擬

Case Study (2): Designing Heterogeneous LLM Agents for Financial Sentiment Analysis

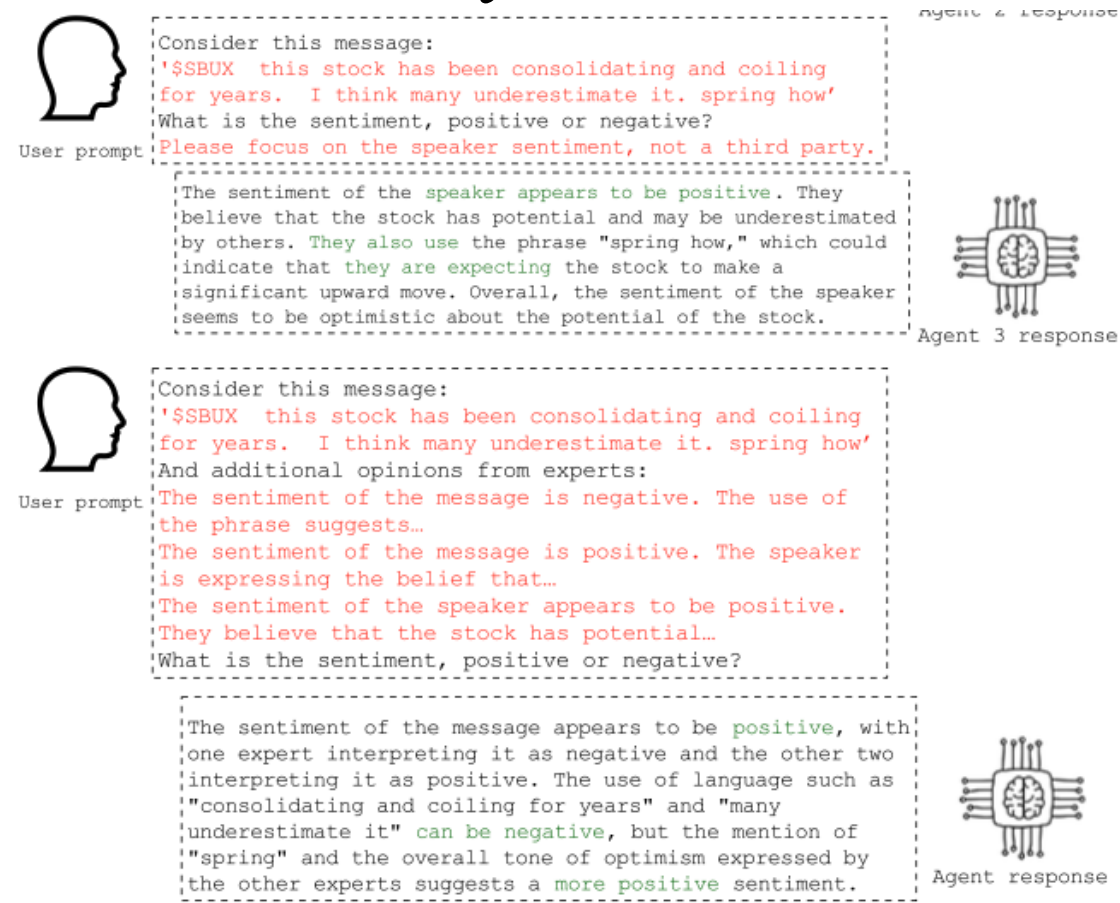


Fig. 3. An illustrative comparison between naïve prompting (the upper example) and the proposed HAD framework (the lower example) with heterogeneous agents inspired by FSA error types. The illustration only shows 3 out of 7 specialized agents due to space limit.

Case Study (2): Designing Heterogeneous LLM Agents for Financial Sentiment Analysis

Table 6. Continued

Case No. & Text	Agent Analysis	Agent Stmt.	HAD Stmt.	True Label
(Case 4) Puravankara’s real estate scheme not a CIS: Sebi	(A1) The sentiment is negative. The use of the word “not” indicates the negation of a positive outcome, and the mention of Sebi suggests that there may be regulatory or legal issues associated with Puravankara’s real estate scheme. There does not appear to be any unrealistic mood used in this message.	Negative	Negative	Negative
	(A2) The phrase “not a CIS” may imply suspicion that Puravankara’s real estate scheme was a CIS (Collective Investment Scheme), which is regulated by SEBI (Securities and Exchange Board of India). The use of the term “scheme” also carries a negative connotation, as it can imply a dubious or deceptive plan. Overall, the message communicates a negative view of Puravankara’s real estate activities in relation to SEBI regulations.	Negative		
	(A3) Neutral. The statement just delivers factual information.	Neutral		
	(A4) Neutral.	Neutral		
	(A5) The sentiment appears to be negative, indicated by the mention of Sebi not considering Puravankara’s real estate scheme as a Collective Investment Scheme (CIS). There may be some regulatory or compliance issues with the scheme, which could have negative implications for the company and its investors. Additionally, the fact that the scheme is being scrutinized by a regulatory authority implies a potential risk or concern. However, without further details about the specific nature of the scheme or the reasons for Sebi’s decision, it is difficult to fully assess the sentiment.	Negative		
	(A6) Cautious and skeptical. The statement suggests uncertainty or controversy surrounding the real estate scheme and its classification. This could raise red flags for institutional investors who seek stability and transparency in their investments.	Negative		
	(A7) Negative. The statement suggests uncertainty or lack of clarity about Puravankara’s real estate scheme. This uncertainty could lead to negative price changes and could negatively impact technical indicators for the company.	Negative		

Case Study (2): Designing Heterogeneous LLM Agents for Financial Sentiment Analysis

Model\Dataset	FPB		StockSen		CMC		FiQA		SEntFiN		FinEntity	
	Acc.	F-1	Acc.	F-1	Acc.	F-1	Acc.	F-1	Acc.	F-1	Acc.	F-1
L & M Dictionary [31]	69.89	54.40	70.70	81.70	–	–	–	–	–	–	–	–
(Fin-)BERT	91.69	89.70	76.90	84.50	93.50	–	–	–	94.29	93.27	–	83.00
BloombergGPT	–	51.07	–	–	–	–	–	75.07	–	–	–	–
BLOOMZ-560m	34.63	32.90	63.65	72.47	87.16	92.62	78.33	83.64	51.32	41.87	32.43	26.90
BLOOMZ-560m (MSV)	29.57	30.13	65.46	74.41	–	–	77.87	82.83	–	–	32.57	27.98
BLOOMZ-560m (MD)	34.25	33.97	68.90	79.46	–	–	78.80	84.21	–	–	32.71	28.37
BLOOMZ-560m (HSV)	33.11	29.01	69.09	83.47	–	–	78.50	83.85	–	–	46.36	22.96
BLOOMZ-560m (HAD)	34.89	38.40	68.06	78.44	87.67	92.95	77.42	83.31	50.16	40.69	34.69	32.93
LlaMa3-70b	45.49	50.00	70.78	71.79	78.25	64.73	84.65	81.44	58.34	52.39	43.21	43.78
LlaMa3-70b (MSV)	42.23	46.87	75.26	84.88	81.47	88.76	85.73	89.38	57.70	50.36	42.75	42.78
LlaMa3-70b (MD)	46.27	51.03	76.35	73.61	77.05	63.00	84.96	84.55	55.36	47.77	45.83	44.69
LlaMa3-70b (HSV)	40.85	40.80	77.67	80.41	79.02	87.39	78.34	84.91	51.73	40.66	40.75	38.22
LlaMa3-70b (HAD)	68.48	84.31	74.25	76.12	71.91	82.06	92.09	94.01	63.51	66.72	60.23	59.36
GPT-3.5	78.58	81.06	67.64	73.93	85.31	91.05	90.53	92.41	67.99	63.21	55.84	56.00
GPT-3.5 (MSV)	72.22	74.68	68.62	76.08	86.31	92.25	87.81	90.29	74.06	64.62	68.46	68.61
GPT-3.5 (MD)	69.83	73.92	69.40	75.72	86.74	91.47	87.95	90.86	69.65	69.30	56.51	57.29
GPT-3.5 (HSV)	65.12	64.95	69.35	76.03	84.04	90.44	87.42	90.05	40.56	64.96	42.37	40.07
GPT-3.5 (HAD)	81.25	87.10	70.01	77.97	90.91	92.69	95.07	96.20	78.16	77.72	61.80	62.56

Conclusion

- Loughran-McDonald Master Dictionary的實證上的效果接近隨機。
- 模型對世界知識的通用推理能力，比單純的領域微調更具 Alpha 價值。
- 架構即：HAD 框架展示了透過軟體架構設計（如代理人角色分配、討論機制）可以模擬人類的高階認知過程。

Thanks